7 Fig. 7.1 shows a circuit containing a 6.0 V battery of cells and three identical resistors.

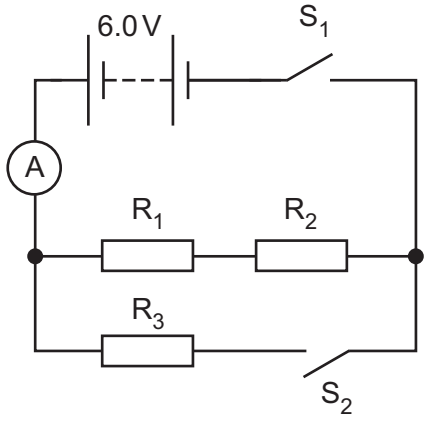

Fig. 7.1

(a) S_1 is closed and S_2 is open. The current in the ammeter is 0.080 A.

Calculate the resistance of R_1.

resistance = ... [2]

(b) S_1 and S_2 are both closed.

(i) Determine the reading on the ammeter. Show your working.

ammeter reading = ... [3]

(ii) Explain in terms of work done and potential difference why there is a larger heating effect in R_3 than in R_1.

.....

.....

..... [2]

[Total: 7]

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